

CS Topics: Mobile Application Development

Orientation Changes, LifeCycles, and Retaining Data

Create an Android app that displays an elf and tracks the elf's mood based on user interactions. Your app must correctly save and restore all of the elf's state when the device rotates, demonstrating your understanding of orientation changes and activity lifecycle methods. Your interface should include:

- an ImageView for the elf's mood
- a TextView describing how the elf feels
- a SeekBar representing the elf's energy level
- a Button that adds cookies for the elf
- a TextView showing the total number of cookies
- a Switch indicating whether the workshop is open

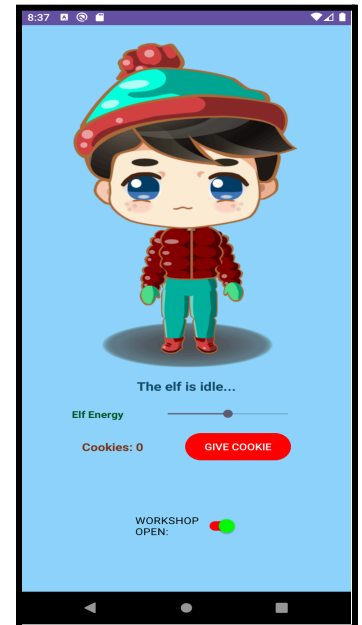


The elf's mood should change as the user interacts with these controls. You may choose the exact rules, but the elf should have different moods, such as: idle, happy, sleepy, concerned, and gifting. For example, high energy might make the elf happy, low energy and very few cookies might make the elf concerned, and an overabundance of cookies might push the elf into gifting mode.

When the screen rotates, all of the following must return exactly as they were before rotation:

- snack count
- SeekBar position
- toggle state
- current elf mood
- the correct mood image
- the mood description text

Your app should use lifecycle logging to see what happens during rotation. Your goal is to demonstrate that you understand how Activities are destroyed, recreated, and how to preserve UI state across configuration changes.



This assignment is intentionally open-ended. Feel free to add extra personality, animations, or holiday cheer, as long as the core functionality works, and the state is restored correctly.

Color Codes: background (#8CD5FA), feeling TextView (#184D6D), cookie TextView (#803515), cookie Button (#FF0000), switch thumb tint (#00FF00), switch track tint (#FF0000), elf energy TextView (#065726)